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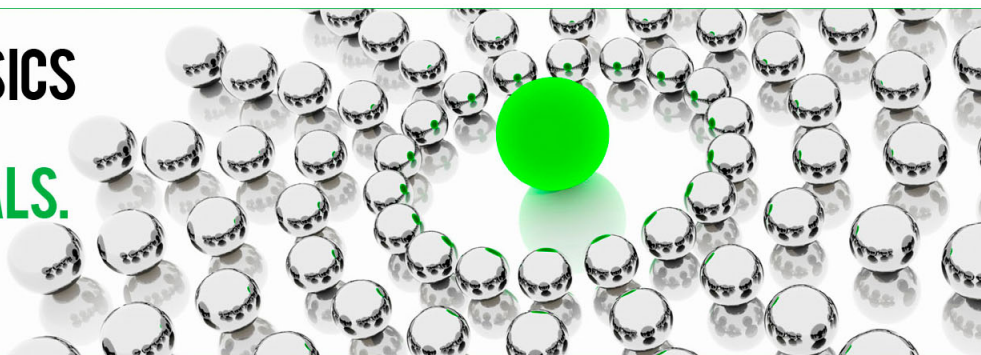
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Specific rate constants $k(E, J)$ for unimolecular bond fissions

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The specific rate constants $k(E, J)$ for unimolecular bond fission reactions are calculated explicitly by a simplified statistical adiabatic channel model. The input data are reactant and product parameters, together with the ratio α/β of a global looseness parameter α and a Morse parameter β characterizing the intermediate part of the potential surface. Angular momentum coupling restrictions are being accounted for. The approach includes rigid RRKM and loose phase space theory as limiting cases for $\alpha/\beta = 0$ and $\alpha/\beta \approx 1$, respectively.

I. INTRODUCTION

In recent years, considerable progress has been made in the direct measurement of specific rate constants of unimolecular reactions. A series of ion fragmentations has been studied successfully, in particular with the PIPECO technique (see, e.g., Refs. 1–4). By laser excitation specific rate constants for isomerization and complex elimination^{5–9} of neutral molecules as well as simple bond fission reactions^{10–12} have also become accessible. With recent experimental progress in this field, energy E —and angular momentum J —selected specific rate constants $k(E, J)$ become measurable for selected sites of initial excitation^{8,12} in the reactant molecule, and with completely resolved state distributions of the product species.^{12,13}

State selected measurements of specific rate constants $k(E, J)$ call for theories which relate the detailed information derived experimentally with the properties of the potential surface of the reaction system. At present, *ab initio* dynamical treatments do not exist even for the simplest unimolecular reactions. Therefore, one has to rely on more conventional approaches. Classical trajectory calculations (see, e.g., Ref. 14) have elucidated a number of aspects of the reaction dynamics. The limitations of this approach can be technical, such as too long trajectories or too large molecules, or more basic, such as various quantum effects. The alternative approach is quantum-statistical theories¹⁵ essentially based on partial equilibrium assumptions similar to transition state theory. These methods replace much of the dynamics by statistics, trading simplicity and practical usefulness against specificity.

In earlier work,¹⁶ we have tried to formulate a most detailed quantum-statistical theory in terms of a statistical adiabatic channel model (hereafter abbreviated SACM). This model includes phase space theory as a limiting case for “loose potential surfaces.” It also includes RRKM theory for “rigid surfaces,” when angular momentum conservation does not pose particular problems, and when state resolved product distributions are not required. For isomerization or complex bond fission reactions with high energy barriers in the forward and reverse directions of the reaction, SACM and RRKM theory become very similar. In this case, “activated complex frequencies,” often guessed in prac-

tical applications, relate to the corresponding saddle point properties of the potential surface. Because of the limited information generally available from both sides, the potential surface calculations and the experiments, no rigorous test of the theory could be made as yet.

For simple bond fission reactions without marked energy barriers for the reverse recombination reaction, conventional RRKM theory has the largest difficulties to account for angular momentum conservation,¹⁷ unless an SACM approach is implemented. The difficulty arises from the transformation of deformation vibrations of the reactant into product rotations during the reaction. Angular momentum coupling plays an important role in this process. In the detailed SACM, these couplings are taken into account individually for all single channel states of the reaction. For this reason, it has been possible to calculate explicitly the J dependence of $k(E, J)$ for simple bond fission reactions.¹⁶ It has also been demonstrated that angular momentum constraints have a marked effect on product energy distributions.^{18–20} In the SACM, for simple bond fissions, the properties of the potential surface have been represented in a simple manner by the reactant and product states, and by an interpolation in terms of the Morse parameter β describing the potential along the “reaction coordinate” and a looseness parameter α characterizing the changes of the other coordinates during the reaction. More detailed representations of channel potentials could be built in when this would appear adequate.

The detailed SACM has been applied to a number of relatively simple molecular systems. With increasing molecular complexity the number of channels increases enormously such that numerical problems may arise. In this case, suitable simplifying techniques appear useful. These simplifications can be made on different levels conserving more or less of the details of the originally completely channel-selected SACM. Some simplifications have already been introduced in the original version of the SACM.¹⁶ More drastic simplifications in terms of a maximum free energy criterion²¹ or a factorization of coordinates²² for the calculation of the thermal high pressure rate constants k_∞ have been described later. It is the aim of the present article to further develop these simplifications for the calculation of specific rate constants $k(E, J)$ in a way that they can be added to conventional RRKM type calculation routines.

II. PRINCIPLE OF COORDINATE SEPARATION IN THE STATISTICAL ADIABATIC CHANNEL MODEL

The basic expression for the calculation of the specific rate constant $k(E, J)$ in statistical theories is

$$k(E, J) = \frac{W(E, J)}{h\rho(E, J)}, \quad (2.1)$$

where $\rho(E, J)$ is the rovibrational density of reactant states and $W(E, J)$ denotes the number of open reactant channels at a given E and J . In RRKM-type transition state theory, $W(E, J)$ is identified with the number of states of one general activated complex of the reaction. At the activated complex, generally all coordinates are separated and represented by harmonic oscillator-rigid rotor approximations. Coupling between the angular momenta of fragment rotations, fragment orbital motion and overall angular momentum J at this point is difficult to account for.²⁰ In the SACM, individual reactant and product states are correlated obeying the relevant angular momentum and symmetry constraints for each channel state. The corresponding channel potential energy curves are constructed and analyzed to find individual activated complexes for all single channel states. In the detailed SACM, all coordinates are always coupled. In our simplification we try to separate the coordinates and to introduce important couplings *a posteriori* by global coupling corrections. Throughout the present work, we shall call all those coordinates the “*disappearing oscillators*” which become product external rotors or orbital motions, whereas we call all other coordinates the “*conserved oscillators*.”

The mildest simplification of the SACM, required for handling large molecules with very large numbers of open channels, consists of separating the groups of disappearing and conserved oscillators. As described in the original version of the SACM,¹⁶ the density of open channels of the disappearing oscillators $\rho_i(E, J)$ is determined first without decoupling, and is convoluted then with the corresponding number of channels for the conserved oscillators $W_r(E_r, J)$ via

$$W(E, J) = \int_0^E W_r(E_r, J) \rho_i(E - E_r, J) dE_r. \quad (2.2)$$

A much more drastic separation of all individual coordinates has been applied in Ref. 22 for the calculation of the thermally averaged high pressure rate constant k_∞ . In the following we further extend this work to the calculation of the specific rate constants $k(E, J)$. The principles of this treatment are the following:

(i) For a given angular momentum J , the lowest open channel determines the threshold energies $E_0(J)$, i.e., the centrifugal barriers.

(ii) For a single disappearing oscillator at $J=0$, the energy pattern of the channel maxima is derived from the SACM. It has been shown in Ref. 22 that the corresponding number of open channel states can be expressed classically by

$$W_{sc} \approx \left(\frac{E - D}{\epsilon} \right)^x \quad (2.3)$$

or quantized by

$$W \approx \text{Int}(W_{sc} + 0.5), \quad (2.4)$$

where $\text{Int}(y)$ denotes the next integer below y and D is the dissociation energy of the Morse potential in the reaction coordinate. Empirical relationships for ϵ and x have been obtained, expressing ϵ and x in terms of the ratio α/β between a “looseness parameter” α and a Morse parameter β characterizing the intermediate part of the potential surface. ϵ and x vary between the corresponding energy quanta ϵ_e and ϵ_∞ (or B_∞) and the exponents x_e and x_∞ of the reactant and product respectively. These empirical relationships are²²

$$x = x_e \exp(-y - y^n) + x_\infty [1 - \exp(-y - y^n)]^n, \quad (2.5)$$

$$y = C_2 \alpha / \beta, \quad (2.6)$$

$$C_2 \approx 1.24 + 55 B_\infty / \epsilon_e, \quad (2.7)$$

$$n \approx 2.25 + (0.5 \times 10^{-3}) \epsilon_e / B_\infty, \quad (2.8)$$

$$\epsilon \approx B_\infty + (\epsilon_e - B_\infty) \exp[-C_3 \alpha / \beta - C_4 (\alpha / \beta)^4], \quad (2.9)$$

$$C_3 \approx 4.00 - 1.29 \ln(\epsilon_e / D), \quad (2.10)$$

$$C_4 \approx 2.80 - 5.19 \ln(\epsilon_e / D). \quad (2.11)$$

For a reactant oscillator, $x_e = 1$; for a product rotor, $x_\infty = 0.5$. For a reactant rotor transforming into a product rotor, $x = x_e = x_\infty = 0.5$. ϵ_e denotes the average of the reactant quanta of all disappearing oscillators (excluding rotational constants), B_∞ denotes the average of all product rotational constants. Equations (2.3)–(2.11) describe the energy pattern of the dynamical barriers of the potential surface, similar to the centrifugal barriers occurring during rotation of the molecules.

(iii) The energy pattern of the channel maxima for conserved oscillators is expressed by a simple interpolation²² of ϵ between the reactant quanta ϵ_e and the product quanta ϵ_∞ :

$$\epsilon \approx \epsilon_\infty + (\epsilon_e - \epsilon_\infty) \exp(-C_3 \alpha / \beta). \quad (2.12)$$

(iv) The energy patterns of the channel maxima for disappearing and conserved oscillators are convoluted to give the total $W(E, J=0)$. Except very close to the threshold energy, the number of open channels for all disappearing oscillators $W_i(E_i, J=0)$ is obtained by integration of the classical expressions (2.3). Convolution with the conserved oscillators is then performed with the Beyer–Swinehart counting routine (see below).

(v) $W(E, J=0)$ is also used for $J \neq 0$ by a simple displacement of the energy scale, i.e., $W(E, J) = 0$ for $E < E_0(J)$. However, a number of important coupling corrections are introduced below.

(vi) The separation of coordinates considered so far violates angular momentum coupling constraints. We correct this error by an angular momentum coupling factor $F_{AM}(E, J)$, which is interpolated analogous to Eq. (2.12), like the quantum of disappearing oscillators, between the reactant and separated fragment expressions which can be calculated explicitly. A similar angular momentum correction factor for factorized partition functions of disappearing oscillators has been used

in Ref. 22 for the calculation of k_∞ [see Eq. (6.7) and Table V of Ref. 22].

(vii) The energy pattern [Eq. (2.3)] of the channel maxima so far has been calculated for $J=0$ and for a single disappearing oscillator being excited. For $J \neq 0$ and for other disappearing oscillators being excited, the effective potential surface for the considered disappearing oscillator changes. In Ref. 22, this was corrected for by modifying the Morse parameter β which is directly related to the well depth of the potential surface. In all α/β relationships then β is replaced by β_{eff} . Since β is inversely proportional to the square root of the well depth, one has

$$\beta_{\text{eff}} \approx \beta \sqrt{\frac{D}{D + [E_0(J) - E_{0\infty}] - B_e J(J+1) - b'[E - B_e J(J+1)]/s}}, \quad (2.13)$$

where $E_{0\infty}$ denotes the threshold energy $E_0(J=0)$ for $\alpha/\beta \rightarrow \infty$, B_e is the rotational constant of the reactant for rotation around an axis perpendicular to the reaction coordinate, s is the number of internal coordinates of the reactant, b' is given by the number of disappearing reactant oscillators minus 1 and excluding "the K rotor" for a nonlinear molecule. Unlike the calculation of k_∞ in Ref. 22 with $\beta_{\text{eff}}(T)$, now one has $\beta_{\text{eff}}(E, J)$.

(viii) Besides $W(E, J)$, the rovibrational density of states $\rho(E, J)$ has to be calculated. Using the Whitten-Rabinovitch approximation this is done below.

(ix) The neglect of symmetry numbers in $\rho(E, J)$ and $W(E, J)$ has to be corrected for. As described in Ref. 22, this can be done by analogy with F_{AM} by dividing by an average symmetry number which is interpolated, analogous to Eq. (2.12), as a function of α/β between

$$B_{\text{cent}}(q) \approx B_e \left\{ \frac{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2}{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q \cos \vartheta_e + m_C(m_A + m_B)q^2} \right\}. \quad (3.2)$$

The threshold energies $E_0(J)$ are then derived numerically from the maxima of Eq. (3.1). It should be noted that in the present work we shall count energies E and $E_0(J)$ from the zero-point level of the reactant molecule such that the zero-point energies included in Eq. (3.1) should be subtracted, see Eqs. (3.2)–(3.4) of Ref. 22. Often it is desirable to provide an analytical representation of the threshold energies. Then the form

$$E_0(J) - E_0(J=0) \approx C_\mu [J(J+1)]^\mu \quad (3.3)$$

proves particularly useful.^{23,24}

B. Rovibrational densities of states $\rho(E, J)$

The rovibrational densities of states $\rho(E, J)$ used for the calculation of $k(E, J)$ are determined by a harmonic oscillator-rigid rotor expression for the reactant molecule $\rho_h(E, J)$, an anharmonicity correction factor $F_{\text{anh}}(E, J)$, and a symmetry number σ_e ,

$$\rho(E, J) \approx \rho_h(E, J) F_{\text{anh}}(E, J) / \sigma_e. \quad (3.4)$$

If necessary, free or hindered internal rotor contributions are included in $\rho_h(E, J)$ (Sec. VIII of Ref. 23).

the corresponding reactant and product values [see Eq. (4.5) in Ref. 22].

The various steps in the calculation of $k(E, J)$ as summarized by (i)–(ix) will be elaborated in detail in the following sections. We shall provide the required expressions for an "exact" fully counted version of the calculation as well as for an approximate "classical" treatment using the Whitten-Rabinovitch relationships.

III. EXPLICIT CALCULATION OF $k(E, J)$

Following the simplifications of the SACM described in the previous section, in this section the specific rate constants $k(E, J)$ are calculated in detail. At first the threshold energies $E_0(J)$ are determined; then the rovibrational densities of states $\rho(E, J)$ are obtained. Finally, the simplified numbers of open channels $W(E, J)$ are derived.

A. Threshold energies $E_0(J)$

Following Sec. V of Ref. 22, the centrifugal barriers $E_0(J)$ are derived numerically from the maxima of the lowest channel potential,

$$V_a(q) = D \{1 - \exp[-\beta(q - q_e)]\}^2 + \Delta E_e \exp[-\alpha(q - q_e)] + E_{\text{sp}} + B_{\text{cent}}(q) J(J+1). \quad (3.1)$$

All symbols are used as in Ref. 22. Often a quasitriatomic rotational energy model gives a sufficient representation of the effective rotational constant $B_{\text{cent}}(q)$, i.e., for a nonlinear triatomic molecular model ABC with a bond angle ϑ_e and equilibrium distances of the A–BC bond q_2 of q_{2e} , and of the AB–C bond q of q_e , one uses

The anharmonicity correction factor in the present work is approximated by calculating $\rho(E, J)$ classically and comparing with the classically determined harmonic density of states

$$F_{\text{anh}}(E, J) \approx \frac{\rho_{\text{sc}}(E, J)}{\rho_{\text{sc}, h}(E, J)}. \quad (3.5)$$

Explicit expressions for $F_{\text{anh}}(E, J)$ are given below.

The harmonic oscillator (–internal rotor) rovibrational densities of states $\rho_h(E, J)$ can be determined by exact state counting techniques (Refs. 25–27 and Appendix A of Ref. 28). However, in general the threshold energy of the reaction E_0 is larger than one-tenth of the reactant zero-point energy, such that the Whitten-Rabinovitch "classical" approximation²⁹ is sufficient. Always omitting the $2J+1$ factor for the M degeneracy of the total angular momentum J states in $\rho(E, J)$ and $W(E, J)$, we express $\rho_h(E, J)$ by

$$\begin{aligned} \rho_h(E, J) &= \rho_{\text{vib}, h}(E - B_e J(J+1)) \\ &\approx \frac{[E - B_e J(J+1) + \alpha(E - B_e J(J+1)E_e)]^{s-1}}{\Gamma(s) \prod_{i=1}^s \epsilon_i} F_p(E, J) \end{aligned} \quad (3.6)$$

with the s reactant quanta ϵ_i , the zero-point energy $E_z = 1/2 \sum_{i=1}^s \epsilon_i$, the Whitten-Rabinovitch function $a(E)$ [see, e.g., Eq. (6) of Ref. 24]. As emphasized above, the energy E is counted from the zero-point level of the reactant molecule. Useful approximate expressions for the degeneracy factor $F_p(E, J)$ are given in Appendix A.

In Eqs. (3.6), (A1)–(A17), the rovibrational densities of states have been calculated neglecting symmetry corrections. Consistent with the determination of $W(E, J)$, we divide $\rho(E, J)$ as given before by the total symmetry number σ_e of the reactant molecule in its equilibrium position including symmetry numbers for internal rotors, see Eq. (3.4).

The calculation of the anharmonicity correction factor $F_{\text{anh}}(E, J)$ from Eqs. (3.4) and (3.5) is not unambiguous. One has to consider metastable molecular states at $E \geq E_0$ for which it is not clear "where the reactant molecule ends." In statistical theories (as well as in trajectory calculations) this question has to be clarified within the basic postulates. One might, e.g., limit the reactant space by the positions of the individual channel maxima. However, this may lead to divergence of $\rho(E, J)$ when such maxima only occur at $q \rightarrow \infty$. Therefore, it appears more reasonable to exclude all states of oscillators above their individual dissociation energies D_i . In our work, we represent the molecule by s Morse oscillators with dissociation energies D_i and obtain the complete $\rho(E, J)$ by convolution. For the energy range of interest in $k(E, J)$ calculations, the corresponding $F_{\text{anh}}(E, J)$ can be factorized approximately in the form

$$F_{\text{anh}}(E, J) \approx \prod_{i=1}^s F_{\text{anh},i}(E, J), \quad (3.7)$$

where $F_{\text{anh},i}$ is the anharmonicity factor for one Morse oscillator with dissociation energy D_i coupled to $s-1$ harmonic oscillators. Classical approximations for $F_{\text{anh},i}$ are given in Appendix B.

C. Number of open channels $W(E, J)$

In the calculation of $W(E, J)$, a number of interpolation relationships are employed as described in Sec. II, and Appendices A and B. These relations use the ratio α/β . The Morse parameter β is calculated in the usual way [Eq. (2.4) of Ref. 22] from the parameters of that reactant oscillator which is identified with the reaction coordinate. The looseness parameter α is put equal to $1 \text{ \AA}^{-1}$ ^{16,18,21} or derived from $\alpha/\beta \approx 0.5$ as "standard values." Later on, α/β remains the only fitting parameter of the present approach to be fixed by comparison with a measured k_∞ or $k(E, J)$ value. Because of the coupling of oscillators, an effective Morse parameter β_{eff} has to replace β as suggested by Eq. (2.13). In principle, β_{eff} depends on E and J such that the determination of $W(E, J)$ requires new sets of quanta and other factors for each energy and angular momentum. Whenever this appears too cumbersome, β_{eff} should be determined for an average E and J of interest for the respective application. It should be noted that only the quanta ϵ and exponents x of the disappearing oscillators sensitively depend on β_{eff} . For all other factors, an average $\alpha/\beta_{\text{eff}}$ appears adequate in any case.

With the quanta ϵ_i and exponents x_i of all conserved and disappearing oscillators (including internal rotors and, e.g., the K rotor for a symmetric top), the individual numbers of open channels $W_i(E, J)$ are convoluted by analogy with Eq. (2.2) to obtain the uncorrected total number of open channels $W^0(E, J)$. For a fully quantized treatment, the detailed Stein-Rabinovitch counting algorithm²⁷ has to be used. However, for energies which are more than a few rotational constants of the fragments above the threshold energies $E_0(J)$, all rotor-type coordinates can be convoluted classically by successive application of convolutions as in Eq. (2.2). In this way, r free internal rotors with rotational constants B_i and l disappearing oscillators with quanta ϵ_j and exponents x_j [see Eq. (2.3)] lead to a classical combined number of states,

$$W_i(E^*) = \frac{[\Gamma(\frac{1}{2})]^r \prod_{j=1}^l \Gamma(1+x_j)}{\Gamma(r/2+1+\sum_{j=1}^l x_j)} \times \prod_{i=1}^r \left(\frac{E^*}{B_i}\right)^{1/2} \prod_{j=1}^l \left(\frac{E^*}{\epsilon_j}\right)^{x_j} \quad (3.8)$$

(see Appendix A of Ref. 28, internal rotor-symmetry numbers are treated separately below). When some of the r internal rotors are not free but hindered, their respective contributions to Eq. (3.8) could be treated approximately, e.g., by Eqs. (3.6)–(3.9) of Ref. 23. As described in Appendix A of Ref. 28, Eq. (3.8) then is quantized and employed for the construction of the starting array $T(I)$ of the most rapid Beyer-Swinehart counting algorithm applied to all remaining conserved harmonic oscillators. This approach represents the fastest presently available algorithm for the accurate and quantized determination of $W(E^*)$. So far, in Eqs. (2.3) and (3.8), the energy E^* has been counted from the classically lowest energy level of the rotor-type coordinates. In order to place $W(E^*)$ at the adequate energy scale to obtain $W^0(E, J)$, we now identify the energy E^* , in $W_i(E^*)$ of Eq. (3.8) and in the then determined total $W(E^*)$, with the energy in excess of the threshold energy, i.e., with $E - E_0(J)$.

Multiplication with angular momentum and symmetry correction factors $F_{\text{AM}}(E, J)$ and $\sigma(E, J)$ to $W^0(E, J)$ completes the calculation of $W(E, J)$:

$$W(E, J) = W^0(E, J) F_{\text{AM}}(E, J) / \sigma(E, J). \quad (3.9)$$

As discussed in Sec. II, F_{AM} and σ are interpolated as the quanta of conserved oscillators [Eq. (2.12)] between the corresponding reactant and product values via

$$F_{\text{AM}}(E, J) \approx F_{\text{AM}\infty}(E, J) + [F_{\text{AM}e}(E, J) - F_{\text{AM}\infty}(E, J)] \exp(-C_3 \alpha / \beta_{\text{eff}}), \quad (3.10)$$

$$\sigma(E, J) \approx \sigma_\infty + (\sigma_q - \sigma_\infty) \exp(-C_3 \alpha / \beta_{\text{eff}}) \quad (3.11)$$

with the effective Morse parameter $\beta_{\text{eff}}(E, J)$. The symmetry number σ_∞ in Eq. (3.11) is given^{21,22} by the product

$$\sigma_\infty = \sigma_1 \sigma_2 S_\infty \quad (3.12)$$

of the total symmetry numbers σ_1 and σ_2 of the two fragments, and $S_\infty = 2$ for identical fragments or $S_\infty = 1$ otherwise. The symmetry number σ_q in Eq. (3.11) is given²¹ approximately by the total symmetry number of

the reactant being distorted along the reaction coordinate q , but otherwise kept in the most symmetrical conformation. The detailed calculation of the reactant and product angular momentum correction factors F_{AMe} and $F_{AM\infty}$ is a somewhat involved task which is performed explicitly in Appendix C.

IV. CLASSICAL EXPRESSIONS FOR SPECIFIC RATE CONSTANTS $k(E, J)$

In Sec. III, the simplified statistical adiabatic channel model has been elaborated in detail allowing for a fully counted determination of specific rate constants $k(E, J)$. Whenever possible, this approach should be chosen in order to always obtain satisfactory results. However, for a large range of energy and angular momentum values, a modified classical Whitten–Rabinovitch approximation also gives good results. For this reason, we provide this alternative formalism in the following. It should, however, be emphasized that this approach may become inapplicable at low excess energies $E - E_0(J)$, see below.

When the threshold energies $E_0(J)$ and the rovibrational densities of states have been determined, the threshold rate constants follow simply from

$$k(E = E_0(J)) = \frac{1}{h\rho(E = E_0(J))}. \quad (4.1)$$

At general energies E and angular momenta J , $k(E, J)$ is calculated by Eq. (2.1) with the density of states as described above from Eqs. (3.4)–(3.7). The number of open channels can be given in a modified Whitten–Rabinovitch approximation. We note that, for $E^* \rightarrow 0$ and for not too small ϵ_i , the classical Whitten–Rabinovitch expression

$$W_{\infty}(E^*) \simeq \frac{(E^* + a(E^*)E_s)^s}{\Gamma(s+1) \prod_{i=1}^s \epsilon_i} \quad (4.2)$$

may approach a value which falls below unity for large s . In order to force $W_{\infty}(E^*)$ to the correct value of unity at $E^* \rightarrow 0$, we have proposed²⁴ simply to add unity at the right-hand side of Eq. (4.2). In order to demonstrate the quality of this “modified Whitten–Rabinovitch expression”

$$W(E^*) \simeq 1 + \frac{(E^* + a(E^*)E_s)^s}{\Gamma(s+1) \prod_{i=1}^s \epsilon_i}, \quad (4.3)$$

in Fig. 1 we compare accurately counted $W(E^*)$ with the results from Eq. (4.3) for a molecule with $s=39$ (cycloheptatriene from Ref. 28). The agreement is very good. It should be emphasized that this modification does not improve the Whitten–Rabinovitch expression when $W_{\infty}(E^*)$ exceeds unity for $E^* \rightarrow 0$. This will often be the case when $\prod \epsilon_i$ includes rotational constants of internal rotors or rotor-type disappearing oscillators. Here, a counting treatment becomes obligatory at low energies E^* .

For the present purpose we go one step further. For $W(E, J)$ we have to consider $l (+r)$ nearly rotor-type “disappearing oscillators” (plus internal rotors) together with s_p “conserved oscillators” (without internal rotors). Below the energies where the conserved oscillators are

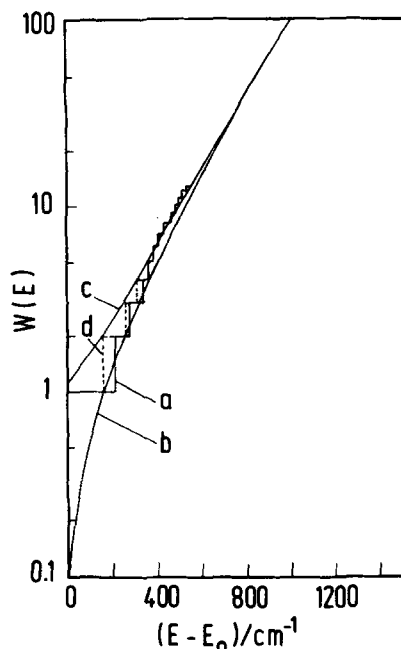


FIG. 1. Number of open channels for a rigid system {isomerization of cycloheptatriene, data from Ref. 28}. Curve a: exact counting; (b): Whitten–Rabinovitch approximation [Eq. (4.2)]; c: modified Whitten–Rabinovitch approximation [Eq. (4.3)]; d: as c but using integer operation.

excitable, Eq. (4.3) would give a very bad representation of the rotor-type coordinates. For this reason, we propose to use

$$W(E, J) \simeq 1 + W_{sc}^{l,r,\Phi}(E, J) F_{AM}^{l,r,\Phi}(E, J) / \sigma(E, J) \quad (4.4)$$

at energies $E - E_0$ smaller than the lowest quantum E^{**} of the s_p product oscillators. $W_{sc}^{l,r}$ and $F_{AM}^{l,r}$ apply to the rotor-type coordinates only. At energies above E^{**} , we use

$$W(E, J) \simeq W_{sc}^{l,r,\Phi}(E, J) F_{AM}^{l,r,\Phi}(E, J) / \sigma(E, J), \quad (4.5)$$

where $W_{sc}^{l,r,\Phi}$ and $F_{AM}^{l,r,\Phi}$ include all coordinates. At energies near E^{**} , there will be a more or less pronounced discontinuity between the two approximations which can only be avoided by the accurate counting procedure. The generalized Whitten–Rabinovitch expressions for $W_{sc}^{l,r}$ and $W_{sc}^{l,r,\Phi}$ are

$$W_{sc}^{l,r}(E, J) \simeq \frac{\{\Gamma(\frac{1}{2})\}^r}{\Gamma(r/2 + 1 + \sum_{j=1}^l x_j)} \times \prod_{i=1}^r \left(\frac{E - E_0(J)}{B_i} \right)^{1/2} \prod_{j=1}^l (1 + x_j) \left(\frac{E - E_0(J)}{\epsilon_j} \right)^{x_j} \quad (4.6)$$

and

$$W_{sc}^{l,r,\Phi}(E, J) \simeq \frac{\{\Gamma(\frac{1}{2})\}^r}{\Gamma(r/2 + 1 + s_p + \sum_{j=1}^l x_j)} \times \prod_{i=1}^r \left(\frac{E^*}{B_i} \right)^{1/2} \prod_{i=1}^{s_p} \left(\frac{E^*}{\epsilon_{\infty i}} \right) \prod_{j=1}^l \Gamma(1 + x_j) \left(\frac{E^*}{\epsilon_j} \right)^{x_j}, \quad (4.7)$$

where $E^* = E - E_0(J) + a^*((E - E_0(J))E_s^*)$ is calculated with the Whitten–Rabinovitch function $a(E)$ and the zero-point energy E_s for the s_p conserved oscillators whose quanta are $\epsilon_{\infty i}$. The exponents x_i and quanta ϵ_i for disappearing and conserved oscillators or internal rotors

are all as described in Secs. II and III. For hindered rotors, again the approximations of Ref. 23 should be incorporated in Eq. (4.7).

V. APPLICATIONS

In this section, we demonstrate the application of the present simplified statistical adiabatic channel model to two model reactions (using realistic molecular parameters) for which explicit calculations by the detailed SACM are also available from Ref. 16. We choose the dissociations $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ and $\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$.

A. $\text{NO}_2 \rightarrow \text{NO} + \text{O}$

For this reaction, $k(E, J)$ values up to $E - E_0(J) = 5500 \text{ cm}^{-1}$ have been calculated in Ref. 16 for $J = 0, 30$, and 60 . At these energies, the NO fragment can be excited in the states $v = 0, 1$, and 2 . Therefore, the convolution of the classical expressions for $W(E, J)$ of the disappearing oscillators with the counted contribution from the single conserved oscillator becomes particularly simple. We specify $k(E, J)$ with respect to v and obtain,

$$k(E, J) = \sum_v k(E, J, v). \quad (5.1)$$

The molecular parameters of Ref. 16 are used again, calculations being performed for $\alpha = 1.3 \text{ \AA}^{-1}$ corresponding to $\alpha/\beta = 0.388$. At first, one obtains the adiabatic zero-point barrier $E_0(J=0) - E_{0\infty} = 2.76 \text{ cm}^{-1}$ and centrifugal barriers represented by Eq. (3.3) with $\nu = 1, 152$ and $C_\nu = 0.029 \text{ cm}^{-1}$ (for the range $J = 0 \rightarrow 120$). $\rho(E, J)$ is calculated with a symmetry number $\sigma_e = 2$ and an anharmonicity factor

$$F_{\text{anh}}(E, J) \approx \left\{ 1 + \frac{1}{3} \left[\frac{D + (E - E_{0\infty}) - J(J+1)B_e}{D + (E_0(J) - E_{0\infty}) - J(J+1)B_e} \right]^3 \right\} \quad (5.2)$$

from Eq. (B8). The Whitten-Rabinovitch harmonic oscillator density of states $\rho_h(E, J=0)$ agrees very well with the accurately counted result. This can be seen in Fig. 2 by the good agreement of the threshold rate constant $k(E_0(J=0))$ from this work and Ref. 16. Nonvanishing threshold rate constants for $J > 0$ are only obtained if the modified classical expression of Eqs. (4.4) and (4.6) is used forcing $W(E, J)$ to unity. The changes of $\beta_{\text{eff}}(E, J)$ with E and J in this example are only small since there are no disappearing oscillators contributing to Eq. (2.13) (i.e., $b' = 0$). The interpolation scheme of Eqs. (2.5)–(2.12), for $J = 0$ leads to quanta ϵ_i and exponents x_i of the disappearing oscillators (corresponding to bending oscillator and K rotor) of $\epsilon_1 \approx 1.84 \text{ cm}^{-1}$, $x_1 \approx 0.50$, $\epsilon_2 \approx 18.3 \text{ cm}^{-1}$, $x_2 \approx 0.58$, to be compared with the rotational constant of the NO rotor of $B_\infty = 1.696 \text{ cm}^{-1}$. One recognizes that the system is far from the phase space limit where $\epsilon_1 = \epsilon_2 = B_\infty$. The angular momentum coupling factor F_{AM} is very small for $J = 0$; essentially it removes one of the two rotor-type disappearing "oscillators" because of the angular momentum constraints (corresponding to $K \leq J$). For $J \geq 30$, F_{AM} approaches $[\Gamma(\frac{3}{2})]^{-2}$, i.e., the two disappearing oscillators fully contribute; however, they are not uncoupled and F_{AM} corrects for this. Correspondingly $W(E, J \geq 30)$ has a steeper rise with energy than $W(E, J = 0)$. Figure 2 demonstrates this effect. Figure 2 also com-

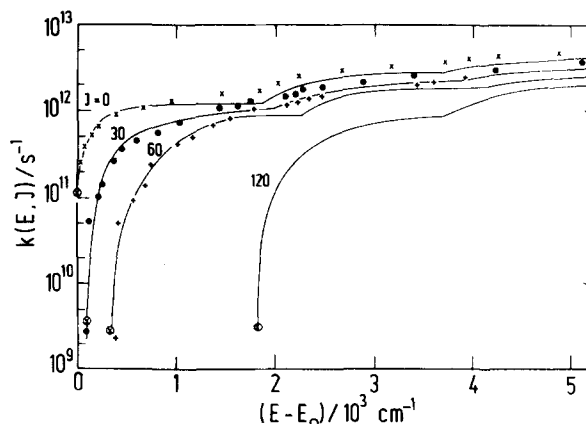


FIG. 2. Specific rate constants $k(E, J)$ for the reaction $\text{NO}_2 \rightarrow \text{NO} + \text{O}$. (Points: SACM of Ref. 16, full lines; this work, using the same parameters as Ref. 16; calculations for $\alpha/\beta = 0.388$).

pares the present $k(E, J)$ values with the results from the detailed SACM.¹⁶ The agreement is satisfactory. Small deviations, quite similar to those noticed in Ref. 22 for k_∞ , originate essentially from the simplified empirical representations of the ϵ_i and x_i of Eqs. (2.5)–(2.12). In practice, they can be compensated in part by minor changes of the anyway empirical parameter α/β . Phase space theory would have produced much larger $k(E, J)$ values. These would correspond to $\alpha/\beta \geq 1$ (see results for $2.1 \text{ \AA}^{-1}$ in Fig. 5 of Ref. 16). The α/β ratio of 0.388 used in Fig. 2 has been fixed to reproduce the experimental k_∞ and $k(E)$ data as indicated in Ref. 16. Therefore, like for all other neutral fragmentations analyzed by us so far, some rigidity of the disappearing oscillators has to be accounted for and introduced into phase space theory, e.g., in the way proposed in the present work. The curves in Fig. 2 also clearly show the "steps" where the NO fragments become excited by one or two quanta. It would be unreasonable to apply a classical Whitten-Rabinovitch treatment to this oscillator.

A number of other observations can be made in Fig. 2. The threshold rate constants $k(E = E_0(J))$ with increasing J at first decrease because of the marked rise of F_p . Later on they increase again because the rise of F_p slows down and is overcompensated by the decrease of the distributable energy $E - B_e J(J+1)$. Whereas F_p at small J rises as $2J+1$, the J -dependence of F_p nearly disappears when the $K \leq J$ constraint becomes unimportant. At small J the $k(E, J)$ curves are not simply shifted along the energy scale according to their different threshold energies $E_0(J)$, whereas for large J they nearly are. The differences are essentially due to the interplay of the various angular momentum constraints.

B. $\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$

For this reaction, $k(E, J=0)$ values up to $E - E_0 \approx 10\,000 \text{ cm}^{-1}$ have been calculated in Ref. 16 using the detailed SACM. We compare our simplified SACM with these results. $k(E, J=0)$, of course, shows a much larger variation with energy than in NO_2 . We have

treated this reaction by accurate counting using the classical number of states $W_1(E^*)$ expression of this work for the construction of the input array $T(I)$ of the Beyer-Swinehart counting routine as described above. These results are shown in Fig. 3 together with a modified classical treatment. C_2H_6 because of its six nearly completely loose disappearing oscillators is a system where the classical treatment breaks down near the threshold energy, since $W_{sc}(E_0) \gg 1$. The situation can be improved very easily by using the modified classical expression of Eqs. (4.4) and (4.6) only. This expression applies nearly exactly at energies below the quanta of the conserved oscillators (i.e., at $E - E_0 \leq 600 \text{ cm}^{-1}$). Above this energy, the classical expression of Eqs. (4.5) and (4.7) can be used. The agreement of the corresponding results with the accurate counting is quite satisfactory. Only very close to the threshold energy E_0 the present results differ from Ref. 16, because symmetry effects now have only been treated by a global symmetry number $\sigma(E, J)$ and not on a single channel basis as in Ref. 16. There is also some graining uncertainty near $E = E_{0\infty}$. This difference so far is of no practical relevance.

VI. CONCLUSIONS

The satisfactory agreement between the present simplified version and the detailed version of the statistical adiabatic channel model has been demonstrated for the NO_2 and C_2H_6 dissociations in Figs. 2 and 3. We expect a similar agreement also for other simple bond fission systems although a more systematic comparison still has to be done. The present approach expresses the specific rate constants $k(E, J)$ in a numerically convenient form in terms of the reactant and product parameters and a single parameter, i.e., α/β , for the intermediate potential surface. This global parameter has to be fixed by comparison with one relevant experimentally measured quantity (although "standard values" of α/β have been observed for several reactions.²¹) Then,

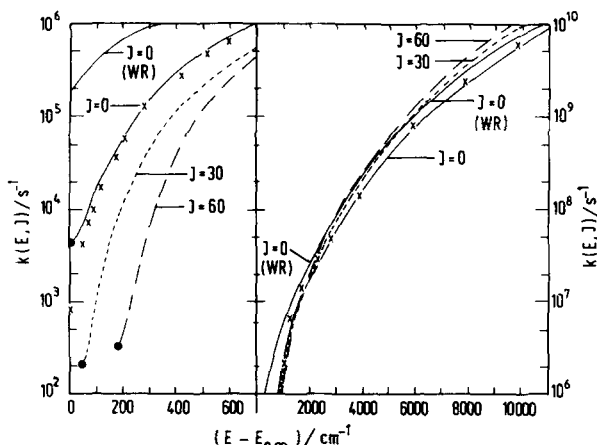


FIG. 3. Specific rate constants $k(E, J=0)$ for the reaction $C_2H_6 \rightarrow 2CH_3$. [Crosses: SACM of Ref. 16, full lines: this work using the same parameters as Ref. 16, calculations for $\alpha/\beta = 0.556$; curve $J=0(WR)$: modified classical approximation of Eqs. (4.4)–(4.7), curve $J=0$: modified classical expression for disappearing oscillators of Eqs. (4.4)–(4.6), crosses: accurate counting for $J=0$.]

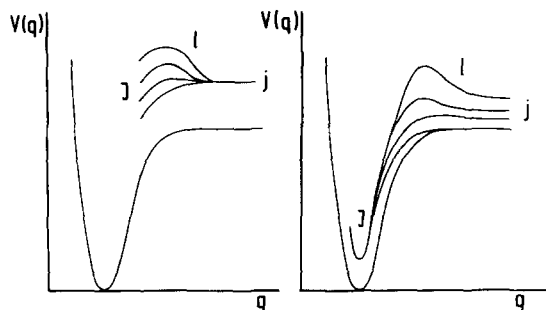


FIG. 4. Schematic representation of centrifugal barriers for atom + linear fragments ($|J-j| \leq l \leq (J+j)$).

the whole range of $k(E, J)$ values can be constructed. By the restriction to the one unknown parameter α/β , the present approach rejects the common practice of RRKM theory of choosing a large set of badly defined "activated complex frequencies." It should, however, be emphasized that "nature cannot be so simple," i.e., that more detailed representations of the intermediate part of the potential surface will be required when sufficiently detailed experiments become available. Also, the relationship between the global "looseness parameter" α and the details of an explicit potential surface calculation remains to become established. Nevertheless, we think that the simplified statistical adiabatic channel model already in its present form provides a minimum number of parameter descriptions of simple bond fission reactions representing the basic physical effects in a satisfactorily transparent form.

ACKNOWLEDGMENTS

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APPENDIX A: DEGENERACY FACTORS $F_p(E, J)$ FOR ROVIBRATIONAL DENSITIES OF STATES

In this Appendix, we give approximate expressions for the degeneracy factors $F_p(E, J)$ in the general relationship (3.6) for rovibrational densities of states $\rho(E, J)$ of linear, prolate symmetrical top and oblate symmetrical top molecules. More exact determinations of $\rho(E, J)$ would require explicit counting. For a spherical top reactant with rotational constants $A_e \approx B_e \approx C_e$, one has Eq. (3.6) with the degeneracy factor $F_p(E, J)$ given by

$$F_p(E, J) = 2J + 1. \quad (A1)$$

For symmetrical top reactant molecules, the calculation of $F_p(E, J)$ is more complicated. Here,

$$\rho_h(E, J) = \sum_{K=0}^J g(K) \rho_{vib,h}(E - E_{rot}(J, K)) \quad (A2)$$

is evaluated in Whitten-Rabinovitch approximation as described in the following. Replacing $\sum g(K)$ by integrals, for a prolate symmetrical top $A_e > B_e \approx C_e$, one has to evaluate

$$\rho_h(E, J) \approx \int_{-J}^{+J*} \rho_{vib, h}(E - B_e J(J+1) - (A_e - B_e)K^2) dK, \quad (A3)$$

where * denotes the additional condition

$$0 \leq (A_e - B_e)K^2 \leq E - B_e J(J+1).$$

This integral gives Eq. (3.6) with

$$F_p(E, J) \approx \sqrt{\frac{E - B_e J(J+1) + a(E - B_e J(J+1))E_s}{A_e - B_e}} \times \int_0^{(A_e - B_e)J^2 / (E - B_e J(J+1) + a(E - B_e J(J+1))E_s)} y^{-1/2} (1-y)^{s-1} dy \quad (A4)$$

for

$$0 \leq (A_e - B_e)J^2 \leq E,$$

and

$$F_p(E, J) \approx F_2 = \frac{\Gamma(\frac{1}{2})\Gamma(s)}{\Gamma(s + \frac{1}{2})} \times \sqrt{\frac{E - B_e J(J+1) + a(E - B_e J(J+1))E_s}{A_e - B_e}} \quad (A5)$$

for $E \leq (A_e - B_e)J^2$ and $B_e J(J+1) \leq E$. For small J , Eq.

(A4) approaches

$$F_p(E, J) \approx F_1 = 2J + 1. \quad (A6)$$

Here, $(1-y)^{s-1}$ is omitted in the integral, and $2J$ has been replaced by $2J+1$ in order to requantize the result. The latter procedure is of crucial importance in order to avoid $\rho(E, J)$ going to zero at $J=0$. A convenient and often sufficient approximation of $F_p(E, J)$ in Eqs. (A4) and (A5) is given by,

$$F_p(E, J) \approx \frac{F_1 F_2}{F_1 + F_2}, \quad (A7)$$

for $0 \leq B_e J(J+1) \leq E$. Equation (A7) includes the case of a spherical top of Eq. (A1). For an oblate symmetrical top $A_e \approx B_e > C_e$ one has to evaluate

$$\rho_h(E, J) \approx \int_{-J}^{+J**} \rho_{vib}(E - B_e J(J+1) + (B_e - C_e)K^2) dK, \quad (A8)$$

where ** denotes the additional condition $(B_e - C_e)K^2 \geq B_e J(J+1) - E$. The resulting expressions are more complicated than for the prolate symmetrical top. The explicit expressions have been given in Ref. 24 [Eqs. (32)–(35)]. They can be simplified by analogy with Eq. (A4) with the abbreviations

$$\rho_1 = (2J+1) \frac{[E - B_e J(J+1) + a(E - B_e J(J+1))E_s]^{s-1}}{\Gamma(s) \prod_{i=1}^s \epsilon_i}, \quad (A9)$$

$$\rho_2 = (2J+1) \frac{[(B_e - C_e)J^2 + a(B_e - C_e)J^2]E_s^{s-1}}{\Gamma(s)(2s-1) \prod_{i=1}^s \epsilon_i}, \quad (A10)$$

$$\rho_3 = \frac{[E - C_e J^2 - B_e J + a(E - C_e J^2 - B_e J)E_s]^s}{\Gamma(s+1) \prod_{i=1}^s \epsilon_i \sqrt{B_e - C_e} \sqrt{B_e J(J+1) - E + a(B_e J(J+1) - E)E_s}}, \quad (A11)$$

by

$$\rho_h(E, J) \approx \rho_1 + \rho_2, \quad (A12)$$

for

$$0 \leq B_e J(J+1) \leq E,$$

and by

$$\rho_h(E, J) \approx \frac{\rho_2 \rho_3}{\rho_2 + \rho_3}, \quad (A13)$$

for

$$(B_e - C_e)J^2 \geq B_e J(J+1) - E \geq 0.$$

Also Eqs. (A9)–(A13) include the case of a spherical top. Equations (A3)–(A13) provide approximations for Eq. (A2) which in general agree with the exactly counted result much better than a factor of 2. When higher precisions are desired, explicit counting following Eq. (A2) becomes necessary.

Linear molecules with degenerate oscillators provide a particular complication. The vibration of the degenerate oscillators generates an angular momentum l which couples with the total angular momentum J . In this case, the restriction $l \leq J$ has to be obeyed. For large J , this restriction becomes irrelevant and $F_p(E, J)$ in Eq. (3.6) is unity. At $J=0$, the degenerate oscillators

have to be omitted from the Whitten–Rabinovitch expression (3.6). As in our treatment of symmetric tops we approximately calculate $F_p(E, J)$ for a linear molecule with one degenerate oscillator. The harmonic rovibrational density of states of $s-2$ nondegenerate oscillators and one doubly degenerate oscillator of quantum ϵ_b in classical approximation is given by the convolution integral

$$\rho_h(E, J) \approx \int_0^{E_{\max}} \frac{(E^* - E')^{s-3} E' dE'}{\Gamma(s-3) \prod_{i=1}^s \epsilon_i}, \quad (A14)$$

where

$$E^* = E - B_e J(J+1) + a(E - B_e J(J+1))E_s$$

and

$$E'_{\max} = \min(J\epsilon_b, E^*).$$

If $E'_{\max} = E^*$, one obtains Eq. (3.6) with $F_p(E, J) = 1$. If $E'_{\max} = J\epsilon_b$ instead, one derives

$$F_p(E, J) \approx \frac{\Gamma(s)}{\Gamma(s-2)} \int_0^{J\epsilon_b/E^*} (1-y)^{s-3} y dy. \quad (A15)$$

By analogy with Eq. (A5), for small J we omit $(1-y)^{s-3}$ from the integral and replace $J^2/2$ by $(1+J)(1+J/2)$ in order to reestablish proper quantization. In this way, we arrive at

$$F_p(E, J) - F_1 = \frac{\Gamma(s)}{\Gamma(s-2)} (1+J)(1+J/2) \left(\frac{\epsilon_b}{E^*} \right)^2. \quad (\text{A16})$$

Combining the two limits as in Eq. (A7), we approximate $F_p(E, J)$ by

$$F_p(E, J) \approx \frac{F_1}{F_1 + 1} \quad (\text{A17})$$

with the general restriction $0 \leq B_e J(J+1) \leq E$.

APPENDIX B: APPROXIMATE ANHARMONICITY FACTORS $F_{\text{anh}}(E, J)$ FOR ROVIBRATIONAL DENSITIES OF STATES

In this Appendix, we give approximate classical expressions for the anharmonicity factors $F_{\text{anh},i}$ of one Morse oscillator of dissociation energy D_i coupled to $s-1$ harmonic oscillators such as factorized in Eq. (3.7). Excluding states of this Morse oscillator above its dissociation energy D_i , the classical number of states is given by²³

$$W_{sc}(E) = \frac{1}{\Gamma(s) \prod_{j=1}^s \epsilon_j} \int_0^{D_i} \frac{(E - E_1)^{s-1}}{\sqrt{1 - E_1/D_i}} dE_1 \quad (\text{B1})$$

for $E \geq D_i$ such that $F_{\text{anh},i}$ follows as

$$F_{\text{anh},i}(E) = (s-1) \frac{D_i}{E} \int_0^1 \frac{(1 - x D_i/E)^{s-2}}{\sqrt{1-x}} dx \quad (\text{B2})$$

and, hence,

$$F_{\text{anh},i}(E) = \left(\frac{s-1}{s-3/2} \right) \frac{D_i}{E} \times \sum_{\nu=0}^{s-2} \frac{\Gamma(s-1)\Gamma(s-3/2-\nu)}{\Gamma(s-3/2)\Gamma(s-1-\nu)} \left(1 - \frac{D_i}{E} \right)^\nu. \quad (\text{B3})$$

For $E \leq D_i$, instead of Eqs. (B1)–(B3), one has

$$W_{sc}(E) = \frac{1}{\Gamma(s) \prod_{j=1}^s \epsilon_j} \int_0^E \frac{(E - E_1)^{s-1}}{\sqrt{1 - E_1/D_i}} dE_1 \quad (\text{B4})$$

and, hence,

$$F_{\text{anh},i}(E) = (s-1) \int_0^1 \frac{(1-x)^{s-2}}{\sqrt{1 - Ex/D_i}} dx \quad (\text{B5})$$

such that

$$F_{\text{anh},i}(E) = (s-1) \sum_{j=0}^{\infty} C_j \frac{\Gamma(1+j)\Gamma(s)}{\Gamma(s+j)} \left(\frac{E}{D_i} \right)^j \quad (\text{B6})$$

with

$$C_0 = 1, \quad C_1 = 1/2,$$

$$C_2 = 1 \cdot 3 / (2 \cdot 4), \quad C_3 = 1 \cdot 3 \cdot 5 / (2 \cdot 4 \cdot 6),$$

etc. Equations (B3) and (B6) adequately describe the increase of $F_{\text{anh},i}$ from unity at $E=0$ to $(s-1)/(s-3/2)$ at $E=D_i$ (see Ref. 23) and its further increase before $F_{\text{anh},i}$ decays toward zero at very high energies. The evaluation of Eqs. (B3) and (B6) shows that this decay is not relevant for calculations of $k(E, J)$. Instead, for all cases which are practically important, Eqs. (B3) and (B6) can be represented with very good precision by an expression linear in E/D_i

$$F_{\text{anh},i}(E) \approx 1 + \frac{E/D_i}{2s-3}. \quad (\text{B7})$$

We have compared the resulting total anharmonicity

factor

$$F_{\text{anh}}(E, J) \approx \prod_{i=1}^s \left[1 + \frac{(E' - B_e J(J+1))/(D_i - B_e J(J+1))}{2s-3} \right] \quad (\text{B8})$$

with exact counts as described above and found excellent agreement. It should be noted that E' is given by the sum of the energy E above the reactant zero-point level and the individual zero-point energy of the oscillator i . In practice there may be some ambiguity about the choice of the various D_i . This uncertainty is of the same order as that of the approximation (B8), and of the general model of separate Morse oscillators.

APPENDIX C: ANGULAR MOMENTUM COUPLING FACTORS $F_{\text{AM}e}(E, J)$ AND $F_{\text{AM}\infty}(E, J)$

The angular momentum correction factors $F_{\text{AM}}(E, J)$, which are applied in the simplified calculation of $W(E, J)$ of Eqs. (3.9), (4.4), and (4.5), are obtained by interpolation between $F_{\text{AM}e}$ and $F_{\text{AM}\infty}$. In this Appendix, these reactant and product expressions are calculated explicitly. $F_{\text{AM}}(E, J)$ is derived by comparing the approximate number of open channel function $W_{\infty}(E, J)$ of Eqs. (4.5)–(4.7) for the reactant-(index e) and product-(index ∞) configurations with the corresponding expressions derived without violating angular momentum coupling restrictions. Since only in the reactant or product configurations is the character of the disappearing oscillators either of pure oscillator or pure rotor type, unique angular momentum coupling schemes can only be designed for these limiting configurations. For intermediate interfragment distances, in the present work the interpolated global angular momentum correction of Eq. (3.10) is employed. Angular momentum coupling schemes sensitively depend on the nature of the species. Therefore, in the following a variety of cases has to be considered.

1. Reactant factors $F_{\text{AM}e}(E, J)$

a. Linear reactant

(i) For the example discussed in Sec. III, the factor $F_{\text{AM}e}(E, J)$ is given by $F_1/(F_1+1)$, where F_1 for numbers of states instead of densities of states by analogy with Eq. (A16) follows by

$$F_1 \approx \frac{\Gamma(s^*+1)}{\Gamma(s^*-1)} (1+J)(1+J/2) \left(\frac{\epsilon_b}{E^*} \right)^2. \quad (\text{C1})$$

s^* and E^* are used as described in the following.

b. Nonlinear reactant

(ii) For a prolate symmetric top reactant of $s-1$ internal coordinates (reaction coordinate omitted), we denote the number of open channels by $W_{\infty}^{(s-1)}(E, J)$. Convoluting $W_{\infty}^{(s-1)}(E, J)$ with the relevant density of states of the K rotor, one obtains

$$W_{\infty}^{(s)}(E, J) \approx \int_{-J}^{+J^*} W_{\infty}^{(s-1)}(E, K, J) dK \quad (\text{C2})$$

with $*$ denoting the additional restriction

$$0 \leq (A_e - B_e)K^2 \leq E - E_0(J).$$

Completely in analogy with Eqs. (A3)–(A5), Eq. (C2) leads to

$$W_{sc}^{(s)}(E, J) \simeq \frac{W_{sc}^{(s-1)}(E, J)(E^*)^{1/2} \Gamma(\frac{3}{2}) \Gamma(s^*)}{\sqrt{A_e - B_e} \Gamma(s^* + 1/2)} \quad (C3)$$

$$\equiv W_{sc}^{(s-1)}(E, J) F_2^*$$

for

$$(A_e - B_e)J^2 \geq E - E_0(J)$$

with

$$E^* = E - E_0(J) + a^*(E - E_0(J))E_2^* \quad (C4)$$

and

$$s^* = 1 + \sum_{i=1}^{s-1} x_{ie} \quad (C5)$$

For

$$0 \leq (A_e - B_e)J^2 \leq E - E_0(J),$$

one obtains

$$W_{sc}^{(s)}(E, J) \simeq \frac{W_{sc}^{(s-1)}(E, J)(E^*)^{1/2}}{\sqrt{A_e - B_e}} \times \int_0^{(A_e - B_e)J^2 / [E - E_0(J) + a^*(E - E_0(J))E_2^*]} y^{-1/2} (1 - y)^{s-1} dy \quad (C6)$$

By analogy with Eq. (A6), Eq. (C6) for small J is approximated by

$$W_{sc}^{(s)}(E, J) \simeq W_{sc}^{(s-1)}(E, J)[2J + 1] \equiv W_{sc}^{(s-1)}(E, J)F_1^* \quad (C7)$$

By analogy with Eq. (A7), a convenient interpolation between the low J and high J expressions (C7) and (C3) is given by

$$W_{sc}^{(s)}(E, J) \simeq W_{sc}^{(s-1)}(E, J) \frac{F_1^* F_2^*}{F_1^* + F_2^*} \quad (C8)$$

The original convolution without obeying angular momentum coupling constraints would have given in Eq. (4.7),

$$W_{sc}(E, J) \simeq W_{sc}^{(s-1)}(E, J) \frac{\Gamma(\frac{3}{2}) \Gamma(s^*)}{\Gamma(s^* + 1/2)} \sqrt{\frac{E^*}{A_e}} \quad (C9)$$

such that the angular momentum coupling correction is

$$F_{AMe}(E, J) = \frac{W_{sc}^{(s)}(E, J)}{W_{sc}(E, J)} \simeq \frac{(2J + 1) \sqrt{A_e / (A_e - B_e)}}{(2J + 1) + \frac{\Gamma(\frac{3}{2}) \Gamma(s^*)}{\Gamma(s^* + 1/2)} \sqrt{\frac{E^*}{A_e - B_e}}} \quad (C10)$$

Equation (C10) includes the spherical top expression

$$F_{AMe}(E, J) \simeq (2J + 1) \frac{\Gamma(s^* + 1/2)}{\Gamma(\frac{3}{2}) \Gamma(s^*)} \sqrt{\frac{A_e}{E^*}} \quad (C11)$$

As before E^* and s^* are given by Eqs. (C4) and (C5). Here, all oscillators are included for $W_{sc}^{(s-1)}$, but only the disappearing oscillators are included for $W_{sc}^{(s)}$.

For an oblate symmetrical top, $F_{AMe}(E, J)$ can be formulated by analogy with Eqs. (A8)–(A13). However, because for simple bond fission reactions $F_{AM}(E, J)$ will always be closer to $F_{AMe}(E, J)$ than to $F_{AMe}(E, J)$, we recommend for simplicity the use of the spherical top

expression (C11) for any nonlinear molecule, with A_e being the rotational constant used in the calculation of $W_{sc}(E, J)$.

2. Product factors $F_{AM\infty}(E, J)$

Product angular momentum coupling schemes have been treated in detail in phase space theory. It was shown^{30,31} that an accurate, general treatment can be done only with numerical techniques. However, from limiting low J and high J expressions convenient approximate interpolations analogous to Eq. (A7) can be designed. In the following we consider various cases.

a. Fragments = atom + linear molecule

(i) The partial number of open channels for disappearing oscillators at a distributable energy E^* and a total angular momentum J is given by

$$W_l(E^*, J) = \sum_j \sum_l 1, \quad (C12)$$

with the constraints

$$|J - j| \leq l \leq J + j$$

and

$$0 \leq B_{\infty} j(j + 1) \leq E^*,$$

where l denotes the quantum number of orbital rotation of the two fragments and j is the rotational quantum number of the linear fragment with rotational constant B_{∞} . $j_{\max}$ is given by the largest j with

$$B_{\infty} j_{\max}(j_{\max} + 1) \leq E^*, \quad (C13)$$

such that

$$W_l(E^*, J) = \sum_{j=0}^J \sum_{l=j-J}^{J+j} 1 + \sum_{j=j_{\max}+1}^{j_{\max}} \sum_{l=j-J}^{J+j} 1 = (2J + 1)(j_{\max} + 1) - J(J + 1), \quad (C14)$$

for

$$0 \leq J \leq j_{\max}(E^*),$$

and

$$W_l(E^*, J) = \sum_{j=0}^{j_{\max}} \sum_{l=j-J}^{J+j} 1 = (j_{\max} + 1)^2 \equiv W_l(E^*, \text{high } J) \quad (C15)$$

for $J \geq j_{\max}(E^*)$.

For the present product species, $W_l(E^*, J)$ can be obtained explicitly in quantized form, whereas this will not be possible for more complex product species. For these more complicated cases, we shall interpolate $W_l(E^*, J)$ between the low J values and the high J values. We can test this procedure for the present atom + linear fragments as well as for the following atom + spheric fragments. For small J , i. e., $J \ll j_{\max}$, Eq. (C14) is approximated by

$$W_l(E^*, J \rightarrow 0) \simeq (2J + 1)(j_{\max} + 1) = (2J + 1)W_l(E^*, J = 0), \quad (C16)$$

whereas for $J \geq j_{\max}$ one uses Eq. (C15). A convenient

interpolation analogous to Eq. (A7) then is given by

$$W_i(E^*, J) \approx \frac{(2J+1)W_i(E^*, J=0)W_i(E^*, \text{high } J)}{(2J+1)W_i(E^*, J=0) + W_i(E^*, \text{high } J)}. \quad (\text{C17})$$

Equation (C17) provides a lower bound of $W_i(E^*, J)$. An upper bound would be provided by

$$W_i(E^*, J) \approx \min\{(2J+1)W_i(E^*, J=0), W_i(E^*, \text{high } J)\}. \quad (\text{C18})$$

The comparison of Eqs. (C17) and (C18) indicates the uncertainty of the two approximations for $W_i(E^*, J)$. As a better approximation one may try the average of Eqs. (C17) and (C18) or, later on, of the corresponding total angular momentum correction factors.

b. Fragments = atom + spherical top

(ii) By analogy with Eq. (C12) $W_i(E^*)$ is now given by

$$W_i(E^*, J) = \sum_j (2j+1) \sum_l 1 \quad (\text{C19})$$

with the constraints

$$|J-j| \leq l \leq J+j$$

and

$$0 \leq B_{\infty} j(j+1) \leq E^*.$$

By analogy with Eq. (C14), one obtains

$$W_i(E^*, J) = (2J+1)(j_{\max}+1)^2 - (2J+1)(J+1)J/3, \quad (\text{C20})$$

for

$$0 \leq J \leq j_{\max}(E^*),$$

and

$$W_i(E^*, J) = (j_{\max}+1)(2j_{\max}/3+1)(2j_{\max}+1) = W_i(E^*, \text{high } J), \quad (\text{C21})$$

for $J \geq j_{\max}(E^*)$. For small J , Eq. (C20) is approximated by

$$W_i(E^*, J) \approx (2J+1)(j_{\max}+1)^2 \equiv (2J+1)W_i(E^*, J=0). \quad (\text{C22})$$

c. Fragments = linear + linear molecules

(iii) For this case, we denote the rotational quantum numbers and rotational constants of the fragments by j_1 , $B_{\infty 1}$, and j_2 , $B_{\infty 2}$, respectively. The angular momenta j_1 and j_2 couple to form j , which couples with the orbital angular momentum l to form J , such that,

$$W_i(E^*, J) = \sum_{j_1} \sum_{j_2} \sum_j \sum_l 1 \quad (\text{C23})$$

with the constraints

$$0 \leq B_{\infty 1} j_1(j_1+1) \leq E^*,$$

$$0 \leq B_{\infty 2} j_2(j_2+1) \leq E^* - B_{\infty 1} j_1(j_1+1),$$

$$|j_1 - j_2| \leq j \leq j_1 + j_2$$

and

$$|J-j| \leq l \leq J+j.$$

Equation (C23) can be evaluated in explicit form. However, the resulting expressions are complicated such that we immediately proceed to the calculation of $W(E^*, J=0)$ and $W(E^*, \text{high } J)$ which, via Eqs. (C17) or (C18), provides convenient approximate results. Using classical expressions by replacing sums by integrals, and putting $j+1 \approx j$, $j_1+1 \approx j_1$, and $j_2+1 \approx j_2$, from Eq. (C23), we derive

$$W_i(E^*, J=0) \approx \frac{2}{3} \frac{E^{*3/2}}{B_{\infty 1} B_{\infty 2}} \times \{\sqrt{B_{\infty 1}} + \sqrt{B_{\infty 2}} - \sqrt{B_{\infty 1} + B_{\infty 2}}\} \quad (\text{C24})$$

and

$$W_i(E^*, \text{high } J) \approx \frac{E^*}{2B_{\infty 1} B_{\infty 2}}. \quad (\text{C25})$$

For general conditions, we use the classical expression

$$W_i(E^*, J) \approx 1 + \frac{(2J+1)W_i(E^*, J=0)W_i(E^*, \text{high } J)}{(2J+1)W_i(E^*, J=0) + W_i(E^*, \text{high } J)} \quad (\text{C26})$$

or the classical analog of Eq. (C18). The classical expressions (C24) and (C25), as well as the classical analogs of Eqs. (C15), (C16), (C21), and (C22) have been derived earlier in the framework of phase space theory.³¹ For our purpose it appears obligatory to use modifications like Eq. (C26) to prevent $W(E^*, J)$ from going to zero for $E^* \rightarrow 0$.

d. Fragments = spherical top + linear molecule

(iv) The spherical top has the rotational constant $B_{\infty 1}$, the linear molecule $B_{\infty 2}$. We evaluate Eq. (C23) with $\sum_{j_1}$ replaced by $\sum_{j_1} (2j_1+1)$. In agreement with Ref. 31, the results are

$$W_i(E^*, J=0) \approx \frac{E^{*2}}{2B_{\infty 1}^{3/2} B_{\infty 2}^{1/2}} \arcsin \sqrt{\frac{B_{\infty 1}}{B_{\infty 1} + B_{\infty 2}}} \quad (\text{C27})$$

and

$$W_i(E^*, \text{high } J) \approx \frac{8E^{*5/2}}{15B_{\infty 1}^{3/2} B_{\infty 2}}. \quad (\text{C28})$$

e. Fragments = spherical top + spherical top

(v) Here one has to evaluate Eq. (C23) with $\sum_{j_1}$ and $\sum_{j_2}$ replaced by $\sum_{j_1} (2j_1+1)$ and $\sum_{j_2} (2j_2+1)$, respectively. In agreement with Ref. 31, the results are

$$W_i(E^*, J=0) \approx \frac{8E^{*5/2}}{15B_{\infty 1} B_{\infty 2} \sqrt{B_{\infty 1} + B_{\infty 2}}} \quad (\text{C29})$$

and

$$W_i(E^*, \text{high } J) \approx \frac{E^{*3}}{6B_{\infty 1}^{3/2} B_{\infty 2}^{3/2}}. \quad (\text{C30})$$

The classical interpolation is given by Eq. (C26).

In order to derive the angular momentum correction factors $F_{A, M\infty}(E, J)$ from the correct partial numbers of open channels $W_i(E^*, J)$ of the disappearing oscillators, two alternatives exist. In case that the conserved oscillators are treated by an exact counting procedure, $W_i(E^*, J)$ can directly be compared with the corresponding factorized expression in Eq. (3.29). Or, for a classical Whitten-Rabinovitch treatment, $W_i(E^*, J)$ has

TABLE I. Parameters C'_i and C''_i for angular momentum correction factors $F_{AM\infty}(E, J)$ [Eq. (C33)].

Case	Fragments	1	C'_i	C''_i
(i)	atom + linear (B_∞)	2	1	$\sqrt{B_\infty}$
(ii)	atom + sphere (B_∞)	3	4/3	$\sqrt{B_\infty}$
(iii)	linear ($B_{1\infty}$) + linear ($B_{2\infty}$)	4	1/2	$\frac{2}{3} \{ \sqrt{B_{1\infty}} + \sqrt{B_{2\infty}} - \sqrt{B_{1\infty} + B_{2\infty}} \}$
(iv)	sphere ($B_{1\infty}$) + linear ($B_{2\infty}$)	5	8/15	$\frac{1}{2} \sqrt{B_{2\infty}} \arcsin \sqrt{\frac{B_{1\infty}}{B_{1\infty} + B_{2\infty}}}$
(v)	sphere ($B_{1\infty}$) + sphere ($B_{1\infty}$)	6	$\pi/6$	$\frac{8}{15} \sqrt{\frac{B_{1\infty} B_{2\infty}}{B_{1\infty} + B_{2\infty}}}$

to be convoluted classically with the respective contributions from the s_p conserved oscillators (and r_p internal product rotors) and again compared with the corresponding factorized expression Eq. (4.7). These two alternative angular momentum correction factors $F_{AM\infty}(E, J)$, as well as those for $F_{AMe}(E, J)$ from Eqs. (C10) and (C11), in the following are combined in one expression; i.e., we give the relevant equation for the Whitten-Rabinovitch treatment, including the case of l disappearing oscillators and r_p internal rotors required for a counting approach by allowing $s_p = 0$ and $a^*(E)E^* = 0$.

We finally have to specify the "distributable energy" E^* . Within the SACM, J states split up into various l states. This is illustrated schematically in Fig. 4. In our present simplified model, we represent the group of l channels from the range $|J - j| \leq l \leq J + j$ by the use of $E_0(J)$ by an average centrifugal barrier. In this spirit, we employ

$$E^* \approx E - E_0(J) \quad (C31)$$

with $E_0(J)$ determined from Eq. (3.1).

The classical convolution of $W_i(E^*, J)$ with the density of states $\rho(E_1)$ of the conserved oscillators is done via

$$W(E, J) \approx \int_0^{E^*} W_i(E^* - E_1, J) \rho(E_1) dE_1. \quad (C32)$$

Comparing the result with the factorized form of Eq. (4.7) via Eq. (C17) gives the final result for the lower bound of $F_{AM\infty}(E, J)$,

$$F_{AM\infty}(E, J) \approx \frac{C'_i \Gamma(1 + l/2)}{\{\Gamma(\frac{3}{2})\}^l} \times \left\{ 1 + \frac{\Gamma(1 + l/2) \Gamma(s_\infty + \frac{l+1}{2}) \sqrt{E^*} C'_i}{\Gamma(\frac{l+1}{2}) \Gamma(s_\infty + 1 + l/2) (2J+1) C''_i} \right\}^{-1}. \quad (C33)$$

Here $s_\infty = s_p + r_p/2$; l is the number of product external rotors; the parameters C'_i and C''_i are tabulated in Table I for the cases (i)–(v) described above. If a counting approach for s_p conserved oscillators is chosen, Eq. (C33) is used with $s_p = 0$ and $E^* = E - E_0(J)$; similarly, then, in Eqs. (C10) and (C11) $E^* = E - E_0(J)$ should be used and s^* be given by the number corresponding to the $l + r_p$ coordinates in the equilibrium position. If on the other hand, the Whitten-Rabinovitch approximation

is used, E^* is given by

$$E - E_0(J) + a^*(E - E_0(J))E^*$$

where a^* and E^* apply to the s_p product oscillators.

By analogy with Eq. (C33) with the approximation (C18), one obtains the upper bound,

$$F_{AM\infty}(E, J) \approx \min \left\{ \frac{\Gamma(1 + s_\infty + l/2) \Gamma((l+1)/2) (2J+1) C''_i}{\Gamma(s_\infty + (l+1)/2) \{\Gamma(\frac{3}{2})\}^l \sqrt{E^*}}, \frac{C'_i \Gamma(1 + l/2)}{\{\Gamma(\frac{3}{2})\}^l} \right\}. \quad (C34)$$

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